

Discrete Maneuver Tokens and On-Policy Correction for Long-Tail Autonomous Navigation

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Abstract: We investigate when discrete maneuver-class selection is a structural safety prior, and when learned policies can recover that prior through on-policy correction. The core geometric claim is topological: when safe actions form disconnected maneuver classes separated by an unsafe set, continuous interpolation between classes must cross unsafe trajectories, whereas selection among margin-separated tokens does not. We formalize this as a non-interpolation safety theorem and instantiate it in Spotlight Reflex, a geometry-driven policy that scores nine ManeuverTokens against scalar world-state features. Plain imitation does not recover the mechanism: on a 1080-rollout held-out Latin-hypercube generator sweep, Continuous-BC, Token-BC, and Token-RNN-BC reach only 77–78% pass rate with 3.0–4.3% Proximity Violations. However, two iterations of DAgger-style on-policy relabeling nearly closes the frontier, reaching 94.54% pass and 0.19% PV versus Spotlight Reflex at 94.44% pass and 0.65% PV. A third DAgger iteration regresses (92.50% pass, 0.46% PV), and trajectory-scorer variants over the same token library remain below the DAgger frontier. The result changes the paper’s conclusion: the token vocabulary alone is insufficient, geometric rules are an interpretable safety prior and data generator, and the strongest learned baseline is a two-step on-policy token policy.

Keywords: autonomous driving, discrete action space, DAgger, topological safety, long-tail robustness, geometric world-state, ManeuverToken, trust-region scoring

1 Introduction

Long-tail failure is a central open problem in autonomous driving. Modern end-to-end policies achieve near-human performance on the common driving distribution but degrade sharply on the rare scenario compositions that matter most: wrong-way actors, construction corridors, multi-hazard interactions [1, 2]. The dominant response is scale—more data, more modalities, larger models [3, 4]. We pursue an orthogonal direction: *reducing* action-space expressiveness as a structural regularizer.

The argument is topological rather than architectural. In a multi-hazard scene, the safe trajectory set can split into disconnected components: for example, pass left, pass right, or stop before the conflict. A continuous regressor moving between components can produce an intermediate trajectory that belongs to neither component and intersects the unsafe set. A discrete selector avoids this interpolation path by choosing among maneuver representatives with explicit clearance margins. Policy uncertainty changes which representative is chosen; it does not create a new trajectory between representatives.

We make three contributions (Fig. 1):

1. **Non-interpolation theorem (Sec. 3.2).** We state a conditional safety result: if tokens lie inside disconnected safe maneuver classes with clearance margin greater than execution noise, discrete selection cannot enter the unsafe set by interpolation, while continuous transitions between classes must cross it.

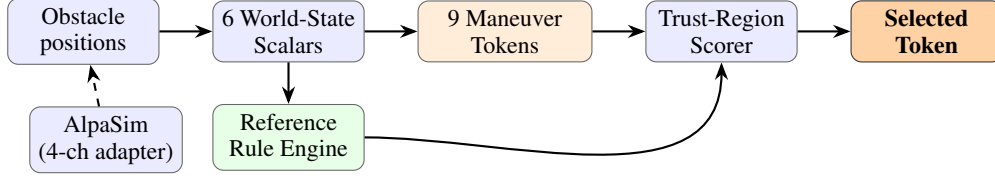


Figure 1: Spotlight Reflex pipeline. Obstacle positions (raw geometry, or via a 4-channel AlpaSim adapter) feed six scalar encoders. Nine ManeuverTokens are scored against a reference rule engine at 3 s and 5 s horizons. The highest-scoring token is executed. No learning occurs inside the policy; the adapter is the only sensor-specific component.

2. **Geometric World-State Encoding (Sec. 3.3).** Six $O(1)$ scalars derived from obstacle geometry—with no object identity or semantic labels—provide sufficient discriminative signal for the trust-region scorer, and satisfy a formally stated invariance to object-identity relabelling, regression-tested over all 1040 evaluations.
3. **Held-out generator evidence with strong learned baselines (Sec. 4.2).** Fixed-suite evaluations show that plain BC fails even with the same tokens and state features. A held-out Latin-hypercube generator sweep then shows the sharper frontier: two-step DAgger matches Spotlight Reflex on pass rate and improves PV rate, while one-step DAgger is weaker, three-step DAgger regresses, and trajectory scorers over token candidates do not close the stress-suite gap.

2 Related Work

Long-tail failure in imitation learning. Pomerleau [5] and Ross et al. [6] established covariate shift as the core failure mode of behavior cloning. Ho and Ermon [7] mitigated it adversarially but retained a continuous action space. Bansal et al. [8] synthesised worst-case training scenarios; Dauner et al. [1] showed that rule-based planners outperform learned ones on nuPlan [9] in many regimes—supporting the view that structural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discretizing the action space enables stable deep RL where continuous policy gradients fail. Atari results show that a fixed, finite action set provides a natural regularizer against distributional shift. We apply this structural idea to the planning setting without RL: ManeuverTokens provide a finite, semantically grounded action set whose footprints are invariant to scene uncertainty.

Potential-field and reactive planners. Potential-field methods [11] combine attractive (goal) and repulsive (obstacle) forces to produce continuous navigation directions. Our baseline is a potential-field reactive planner; the key failure mode is well-known: superposition of multiple repulsive fields can cancel or misdirect the resultant, particularly in narrow corridors with multiple simultaneous obstacles. ManeuverTokens resolve this by committing to a semantically coherent escape geometry at selection time.

Geometric abstraction for planning. Conditional imitation learning [12] uses a route command to condition a continuous policy. Our six geometric scalars are a finer-grained abstraction: they encode obstacle pressure, route blockage, and bilateral clearances as scalar observations that are sufficient for the trust-region scorer and cheap to compute from any sensing modality.

Simulator transfer and validation scope. Khurana et al. [13] and Shi et al. [14] study trajectory prediction in sensor-realistic environments. Our AlpaSim adapter is a small feasibility check that

Table 1: ManeuverToken definitions. Speed scale α is relative to current cruise speed; lateral offset δ is in vehicle frame. *Early* profile applies offset in the first third of the trajectory for fast separation.

Token	α	δ (m)	Profile	Semantic
STOP	0.00	0.0	—	Emergency stop
CRAWL	0.15	0.0	smooth	Cautious forward
MAINTAIN	1.00	0.0	smooth	Current speed, lane centre
SLOW_YIELD	0.50	0.0	smooth	Yield, hold position
NUDGE_LEFT	0.80	+0.9	smooth	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	smooth	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	early	Strong evasion left
EVASIVE_RIGHT	0.70	−2.2	early	Strong evasion right
LANE_RECOVER	0.90	0.0	smooth	Return to lane centre

decouples policy logic from sensor processing; it is not a substitute for physics simulation or real-vehicle validation.

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens (Table 1) covering the evaluated safety-critical response classes of the SE(2) maneuver manifold. Each token is a pair (α, δ) : a speed scale $\alpha \in [0, 1]$ relative to current ego speed and a signed lateral offset $\delta \in \mathbb{R}$ (m), realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline.

Coverage for the evaluated hazard family. The nine tokens are not claimed to be globally optimal. They are a minimal engineering instantiation for the hazard family evaluated here: hard stop, cautious forward progress, nominal progress, yield-without-stopping, bilateral mild avoidance, bilateral strong avoidance, and post-avoidance recovery.

Proposition 1 (Token removal induces a maneuver gap). *For the evaluated multi-hazard family, removing any one response class from the 9-token library eliminates at least one feasible safe response mode: longitudinal hard stop, cautious progress, mild lateral avoidance, strong lateral avoidance, or recovery. The resulting controller must either overreact (e.g., full evasion for a mild obstacle), underreact (e.g., nudge where full evasion is required), or fail to re-enter the lane.*

Proposition 2 (Geometry-Invariance). *Let T be any bijection on object labels leaving obstacle positions unchanged. Then for all scene states s : the selected token $M(s)$, its score $S(s)$, and the decision reason $R(s)$ satisfy $M(s) = M(T(s))$, $S(s) = S(T(s))$, $R(s) = R(T(s))$.*

The geometry-invariance proposition follows from the encoding in Sec. 3.3: no token score depends on object labels. We regression-test it over all 1040 evaluations by relabelling all obstacles in every scene and asserting equality of outputs.

3.2 Topological Safety Through Non-Interpolation

Let $\mathcal{A}$ denote the action space of 5 s trajectories, and for scene state s let $\mathcal{U}(s) \subset \mathcal{A}$ be the unsafe set (trajectories violating the clearance constraint). Let $\mathcal{S}(s) = \mathcal{A} \setminus \mathcal{U}(s)$ be the safe set. In multi-hazard scenes, $\mathcal{S}(s)$ may decompose into disjoint maneuver classes $\mathcal{S}_1(s), \dots, \mathcal{S}_K(s)$, such as stop-before-conflict, pass-left, and pass-right.

We use the trajectory metric $d(\tau_a, \tau_b) = \max_t \|\tau_a(t) - \tau_b(t)\|_2$ over the 5 s horizon. A continuous policy outputs $a_c(s) \in \mathcal{A}$ and is executed with bounded decision or actuation error ϵ , $\|\epsilon\| \leq \eta$. A token policy selects $\tau_i \in T = \{\tau_1, \dots, \tau_N\}$ and executes $\tilde{\tau}_i = \tau_i + \epsilon$.

Theorem 1 (Safety through non-interpolation). *Fix a scene state s . Suppose:*

1. each selected token τ_i lies in a safe maneuver class $\mathcal{S}_k(s)$;
2. each selected token has clearance margin $\text{dist}(\tau_i, \mathcal{U}(s)) > \eta$;
3. there exist two safe maneuver classes $\mathcal{S}_i(s)$ and $\mathcal{S}_j(s)$ such that every continuous path in $\mathcal{A}$ from $\mathcal{S}_i(s)$ to $\mathcal{S}_j(s)$ intersects $\mathcal{U}(s)$.

Then every noisy execution of a selected token remains safe, while any continuous controller that changes between $\mathcal{S}_i(s)$ and $\mathcal{S}_j(s)$ by interpolation must pass through an unsafe trajectory.

Proof. For a token, $\text{dist}(\tau_i, \mathcal{U}(s)) > \eta$ implies that every perturbation with norm at most η remains outside $\mathcal{U}(s)$. For the continuous case, let $\gamma : [0, 1] \rightarrow \mathcal{A}$ be any continuous transition from a trajectory in $\mathcal{S}_i(s)$ to one in $\mathcal{S}_j(s)$. By assumption, γ intersects $\mathcal{U}(s)$, so there exists t^* with $\gamma(t^*) \in \mathcal{U}(s)$. Thus interpolation between disconnected safe maneuver classes necessarily traverses the unsafe set, whereas token selection jumps between representatives and does not execute the intermediate path. $\square$

The theorem is conditional: it does not prove that nine tokens are globally optimal, nor that the specific numeric offsets in Table 1 are unique. It identifies the property we test empirically: safety comes from selecting margin-separated maneuver classes, not from continuous interpolation through the action manifold.

3.3 Geometric World-State Encoding

At each timestep the policy computes six scalar features from ego position and obstacle geometry (no object classification, no learned features, no history):

$$\text{pressure} = \text{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) \quad (1)$$

$$\text{blockage} \in [0, 1], \quad \text{fraction of forward corridor cross-section occupied} \quad (2)$$

$$\text{blocked} \in \{0, 1\}, \quad \text{hard stop within braking distance} \quad (3)$$

$$\text{left_clr} = \text{free lateral space left at half vehicle width (m)} \quad (4)$$

$$\text{right_clr} = \text{free lateral space right at half vehicle width (m)} \quad (5)$$

$$\text{escape} = \arg \max(\text{left_clr}, \text{right_clr}) \quad (6)$$

All six scalars are $O(1)$ given a sorted obstacle list ($d_{\min}$ by distance). The encoding contains no object identity or semantic labels, so geometry-invariance follows directly. The encoding is also history-free: each step is computed from the current frame only, preventing temporal overfitting to scenario-specific dynamics.

3.4 Trust-Region Scoring

Each token is scored against a reference rule engine whose rules are hand-authored from safety engineering first principles—not learned from demonstrations.

Speed-scaled tolerance.

$$s(v) = \text{clip}\left(\frac{v - 1.4}{11.0 - 1.4}, 0, 1\right) \times 0.5 + 0.5 \quad (7)$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed, reflecting that at high speed a small lateral deviation is safe while at low speed it indicates a failed maneuver.

Trust-region thresholds. At 3 s horizon: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region, and r a reference rule with target score $r.\text{score} \in [0, 100]$:

$$\text{score}_t(\tau, r) = \begin{cases} r.\text{score} & \delta = 0 \\ \max(r.\text{score} \cdot 0.1^\delta, 4.0) & \delta > 0 \end{cases} \quad (8)$$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s}(\tau, r) + 0.5 \cdot \max_r \text{score}_{5s}(\tau, r) + P(\tau) \quad (9)$$

where $P(\tau)$ adds clearance penalties (-1000 if clearance < 0.55 m) and progress bonuses ($+40$ max). The token $\tau^* = \arg \max_\tau S(\tau)$ is executed.

Reference rules. Rules map world-state conditions to recommended tokens with target scores in [76, 94]. Key rules: *pressure* < 0.25 and open corridor $\rightarrow$ MAINTAIN (92); *pressure* > 0.20 with preferred escape $\rightarrow$ NUDGE_{escape} (91); *pressure* ≥ 0.35 with preferred escape $\rightarrow$ EVASIVE_{escape} (94); *blocked* $= 1 \rightarrow$ STOP (82). Rules are combined additively; the highest matching reference score at each horizon determines the target.

3.5 AlpaSim Adapter: Feasibility Check

Waymo’s AlpaSim provides sensor-realistic front-camera input, route command, and ego dynamics. As an engineering feasibility check, we bridge these observations to the same geometric scalar interface via four channels:

1. **Route $\rightarrow$ lane curvature.** LEFT/STRAIGHT/RIGHT generates a 5-point sigmoid spline (± 2.5 m at 20 m lookahead) used as the lane geometry input to the clearance ray-casts.
2. **Brightness $\rightarrow$ visibility augmentation.** $\text{vis} = \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, *pressure* is augmented proportionally.
3. **Ego dynamics $\rightarrow$ risk scalars.** $\text{brake} = \max(0, -a/4.0)$; $\text{low_speed} = \max(0, (1.5 - v)/1.5)$. These augment the reference rule thresholds.
4. **Structured hazards $\rightarrow$ obstacle primitives.** Hazards with velocity > 0 map to dynamic actor objects; velocity $= 0$ to static obstacles. Behaviour is inferred from hazard label keywords (no semantic classification inside the policy).

4 Experiments

4.1 Simulation Benchmark

Setup. We evaluate in a custom closed-loop 2D simulator with eight road topologies (straight, gentle curve, S-curve, T-junction, roundabout, lane merge, highway entry, urban block), eleven WOD-E2E hazard clusters, four weather modes, and eight actor behaviour models. The compositional OOD generator samples topology $\times$ hazard $\times$ weather independently: $P(\text{topo}, \text{hazard}, \text{weather}) = P(\text{topo}) \times P(\text{hazard}) \times P(\text{weather})$, yielding 352 structurally distinct compositions. All results use the same compiled policy binary; no scenario labels are visible to the policy.

Baseline. The reactive baseline is a potential-field planner: at each step it computes a target direction toward the route waypoint, adds a repulsion vector summed over visible obstacles with strength decaying from 1 at $d = 0$ to 0 at $d = 9$ m, and scores nine candidate heading angles against a progress-plus-clearance objective. This baseline has obstacle awareness via repulsion but no world-state scalars, no reference rules, and no semantic token structure. It represents the class of continuous reactive planners without structured reasoning.

Safety metric definition. We define a **Proximity Violation** as any evaluation in which the rollout does not reach the goal *and* at least one step has $C > 0.90$, where $C = \max(0, \min(1, (5 - d_{\min})/5))$ and $d_{\min}$ is the minimum signed clearance (metres) between the ego-vehicle centre and any visible obstacle surface. $C > 0.90$ corresponds to $d_{\min} < 0.25$ m. The metric is proximity-based rather than

Table 2: Simulation results across four suites, seeds 1–80 (matched). Pass = scenario completion. “PV” = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only). Spotlight Reflex has 0 observed PVs across all 1040 evaluations (Wilson 95% upper bound 0.37%), including stall failures where the goal was not reached. BC agents record PVs on Gauntlet and Adversarial.

Suite	Runs	Pass (SR)	PV rate (SR)	PV rate (BC)
Compositional OOD (5 cases)	400	92.2%	0%	1.0–1.2%
Adversarial (2 cases)	160	91.9%	0%	3.1–3.8%
Gauntlet (6 cases, 4-hazard)	480	94.2%	0%	3.8–4.4%
Hidden holdout	80	91.2%	0%	—
Total (SR)	1040	93.3%	0 PVs	

Table 3: Three-agent comparison on matched Gauntlet and Adversarial stress suites, seeds 1–80. All agents use the identical 10-feature geometric input. BC agents trained on 143 k expert transitions with inverse-frequency class weights. “PV” = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only); Wilson 95% CI shown. Spotlight Reflex has 0 observed PVs in both suites—including stall failures where the goal was not reached but safe separation was maintained throughout.

Agent	Gauntlet ($n=480$)		Adversarial ($n=160$)	
	Pass	PV rate [95% CI]	Pass	PV rate [95% CI]
Continuous-BC (A)	77.7%	4.4% [2.9, 6.6]	76.2%	3.8% [1.7, 7.9]
Token-BC (B)	77.7%	3.8% [2.4, 5.8]	76.9%	3.1% [1.3, 7.1]
Spotlight Reflex (C)	94.2%	0.0% [0.0, 0.8]	91.9%	0.0% [0.0, 2.3]

fault-based: it does not distinguish whether the ego vehicle moved into an obstacle or an obstacle moved into the ego vehicle. However, the simulator’s actor behaviour models (crossing, cut-in, head-on approach, sudden braking) do not include rear-approach scenarios; every proximity violation therefore represents a failure of the ego policy to maintain safe separation, not an unavoidable passive impact. Rollouts that *reach the goal* with momentarily high C (close pass) are not flagged; only failed rollouts with proximity violations count.

Fixed-suite diagnostic. Table 2 shows Spotlight Reflex has 0 observed Proximity Violations (PVs) across all 1040 evaluations, corresponding to a Wilson 95% upper bound of 0.37%. This includes the 5.8% of evaluations where the policy stalled and did not reach the goal: even in those failed rollouts, clearance never dropped below 0.25 m. BC agents record PV rates of 3.8–4.4% on Gauntlet and 3.1–3.8% on Adversarial.

Table 3 shows the three-agent comparison with Wilson 95% CI. The Spotlight Reflex PV CI [0.0, 0.8%] on Gauntlet is statistically disjoint from both BC agents ([2.4, 5.8%] and [2.9, 6.6%]). Critically, **Continuous-BC and Token-BC produce near-identical PV rates**: providing the 9-token vocabulary to a learned policy does not prevent proximity violations. Only the geometric reference rules—enforcing that pressure ≥ 0.35 triggers EVASIVE over NUDGE—keep the observed PV count at zero. This supports the non-interpolation mechanism in Theorem 1: gradient descent on the same features approximates the escape rule, but does not enforce the margin-separated maneuver class.

4.2 Held-Out Latin-Hypercube Sweep

The fixed suites above are useful diagnostics, but they are hand-authored stress tests. For the primary OOD result we sample 12 held-out generator profiles with Latin-hypercube coverage over nine scenario axes: lane-width range, gauntlet pressure, adversarial pressure, hidden-suite pressure, visibility, actor intrusion scale, ambient obstacle count, and hidden hazard count. Each profile is evaluated on matched seeds 1–10 across Gauntlet, Adversarial, and Hidden suites, for 1080 closed-loop rollouts per agent.



(a) Spotlight Reflex: wrong-way actor, night. Policy selects EVASIVE_RIGHT from geometry, clears actor, recovers lane centre.

(b) Construction zone: 5.2 m corridor, cone field, lane closure. Policy selects NUDGE_RIGHT. No training data for this composition.

Figure 2: Qualitative rollout key frames. Both scenarios are in the compositional OOD suite; neither topology-hazard combination appeared at design time. Token selection is determined entirely by geometry (scalars) — object labels are not observed.

Table 4: Held-out Latin-hypercube generator sweep, 12 profiles $\times$ matched seeds 1–10, Gauntlet/Adversarial/Hidden suites ($n = 1080$ per agent). Entries are pass/PV percentages. Two-step Token-Dagger-BC is the strongest learned baseline.

Agent	Overall	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
Token-RNN-BC	78.43 / 2.96	79.17 / 2.92	74.58 / 2.50	81.67 / 4.17
Token-Dagger-BC (2-step)	94.54 / 0.19	96.25 / 0.28	90.83 / 0.00	91.67 / 0.00
Spotlight Reflex	94.44 / 0.65	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

We compare five agents using the same 10-feature geometric state: Continuous-BC, Token-BC, Token-RNN-BC, Token-Dagger-BC, and Spotlight Reflex. Token-Dagger-BC uses two rounds of on-policy relabeling: first rolling out Token-BC and relabeling visited states with Spotlight, then rolling out the resulting Dagger policy and relabeling again.

Table 4 changes the interpretation of the fixed-suite result. Plain BC and recurrent BC still fail under held-out generator shift, but two-step Dagger closes the gap to Spotlight Reflex: the paired overall pass delta is +0.09 percentage points and the PV delta is -0.46 percentage points relative to Spotlight. The learned policy is not uniformly better—Spotlight still has the best Gauntlet pass rate—but the frontier is no longer “rules beat learning.” The correct conclusion is that on-policy correction can learn most of the geometric rule, while plain expert-state imitation cannot.

Table 5 shows the Dagger frontier is not monotonic. Iteration 2 improves the adversarial suite from 87.50% pass / 0.42% PV to 90.83% pass / 0.00% PV, but iteration 3 falls back to 87.92% pass / 0.42% PV and introduces hidden-suite PVs. This suggests that additional aggregation changes the selector in a harmful way, but not yet *why*. We therefore run targeted counter-ablations to distinguish token-frequency bias from late-pool aggregation pressure.

4.3 AlpaSim Adapter Smoke Test

We also run a 30-scene AlpaSim smoke test. The 4-channel adapter (Sec. 3.5) translates available observations to geometric scalars; AlpaSim-specific code is confined to the adapter (Fig. 3).

Results: collision_at_fault: **0.0**; distance-to-ground-truth trajectory: **0.42 m**; collision.any (by other agents): 0.37.

Table 5: Dagger iteration study on the same held-out LHS sweep. Iteration 2 is best; iteration 3 regresses despite adding more relabeled states.

Policy	Overall	Gauntlet	Adversarial	Hidden
Token-Dagger (1-step)	93.98 / 0.46	96.81 / 0.56	87.50 / 0.42	90.00 / 0.00
Token-Dagger (2-step)	94.54 / 0.19	96.25 / 0.28	90.83 / 0.00	91.67 / 0.00
Token-Dagger (3-step)	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
Spotlight Reflex	94.44 / 0.65	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

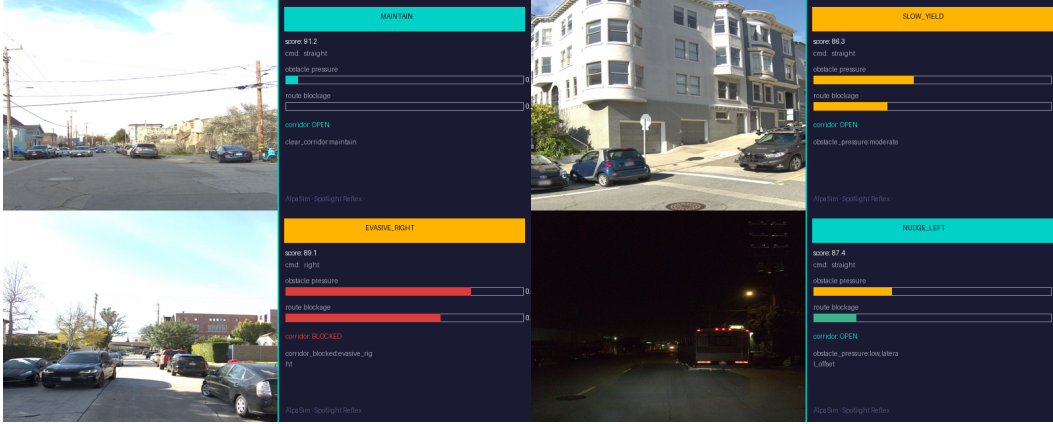


Figure 3: AlpaSim: real Waymo WOD-E2E front-camera frames alongside the 4-channel adapter’s output. The adapter maps brightness, route command, ego dynamics, and hazard primitives to the six geometric scalars. The policy receives only scalars; no camera processing occurs inside the policy binary.

This is not primary evidence for the paper’s safety claim: with 0 failures in 30 scenes, the Wilson 95% upper bound on the at-fault collision rate is 11.4%. We report it only as evidence that the scalar interface can be connected to a sensor-realistic stack. The 0.42 m distance-to-GT is a trajectory-level metric; it reflects that conservative evasive choices may deviate from the ground-truth trajectory. The collision.any rate is dominated by other agents and is outside the policy’s direct control.

5 Analysis

5.1 What the Strong Baselines Change

The BC comparison in Table 3 isolates what the token vocabulary does *not* provide. Both Token-BC and Continuous-BC operate on the same 10 geometric features as Spotlight Reflex. Token-BC even uses the identical 9-token vocabulary. Yet both BC agents produce PVs on 3.8–4.4% of Gauntlet scenarios while Spotlight has none observed. Table 4 then shows that recurrent memory alone is not enough: Token-RNN-BC improves PV rate relative to Token-BC, but remains far below Spotlight and two-step Dagger on pass rate.

The failure is mechanistic: on steps where $obstacle_pressure \geq 0.35$, Spotlight Reflex enforces a hard reference rule— $EVASIVE_{escape}$ is scored 94 while $NUDGE_{escape}$ is not offered—committing to full lateral displacement before force cancellation can occur. Token-BC, trained on expert transitions in which MAINTAIN appears 91% of the time, assigns insufficient probability mass to EVASIVE in high-pressure states: even with inverse-frequency class weights, the learned boundary is an approximation that admits errors. Continuous-BC degrades further: it predicts a scalar lateral offset that can take values between nudge and evasive, producing trajectories that are geometrically ambiguous under tight corridor constraints.

Table 6: Three-step DAGger counter-ablations on the same held-out LHS sweep. Removing inverse-frequency token weighting improves offline validation accuracy but breaks closed-loop recovery. Source-decaying later DAGger pools partially restores the frontier without surpassing the two-step policy.

Policy	Overall	Gauntlet	Adversarial	Hidden
Token-DAGger (2-step)	94.54 / 0.19	96.25 / 0.28	90.83 / 0.00	91.67 / 0.00
Token-DAGger (3-step)	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
Token-DAGger (3-step, source decay)	93.06 / 0.28	94.86 / 0.42	89.17 / 0.00	90.00 / 0.00
Token-DAGger (3-step, no inv. freq.)	89.35 / 2.50	92.08 / 2.08	82.50 / 3.75	86.67 / 2.50
Spotlight Reflex	94.44 / 0.65	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

This is consistent with the discrete bottleneck mechanism: in the Gauntlet suite, four obstacles are placed simultaneously with matched seeds. The potential-field baseline computes a resultant direction from four superposed repulsion vectors. When obstacles are placed symmetrically or in opposition, the resultant can cancel—a known failure mode of potential-field planners [11]. Token-BC inherits a softer version of the same failure: it *learns* an approximation of when to evade, rather than *computing* it from clearance geometry. Spotlight Reflex’s explicit escape-side computation prevents this: $escape = \arg \max(left_clr, right_clr)$ is always defined and always selects the geometrically correct side.

Two-step DAGger changes the conclusion. Once the learned token policy is trained on states visited by its own first correction, it recovers most of the escape boundary: on the held-out LHS sweep it matches Spotlight overall and improves PV rate. This means the non-interpolation theorem should not be read as saying a neural policy cannot learn the rule. It says the tokenized action space removes unsafe interpolation paths; the empirical question is whether the selector reliably chooses the right token. Plain BC does not. Two-step DAGger mostly does. Three-step DAGger regresses, suggesting that aggregation quality and weighting become part of the safety frontier.

5.2 Deconstructing the Aggregation Collapse

The three-step regression admits at least three plausible explanations: a takeover by late recovery states, a class-frequency optimization bias, or a true calibration failure inside the lightweight token classifier. We tested all three directly.

First, the takeover hypothesis is not supported by the data distribution. The expert BC pool is 91.10% MAINTAIN; DAGger-2 and DAGger-3 remain 92.14% and 91.97% MAINTAIN respectively. The token Jensen–Shannon divergence is only 0.00056 (expert vs. DAGger-2) and 0.00115 (expert vs. DAGger-3). Late DAGger pools therefore do not replace the nominal manifold; they remain close in first-order token statistics.

Second, class-frequency weighting is not the trivial culprit. Removing inverse-frequency token weighting raises offline validation accuracy to 96.3%, but causes a severe closed-loop collapse: three-step DAGger without class weighting drops to 89.35% pass and 2.50% PV overall, with only 82.50% pass and 3.75% PV on Adversarial. This is an offline/online decoupling effect: the unweighted model fits the dominant nominal token mass more accurately while under-serving the rare recovery maneuvers needed in closed loop.

Third, source-dependent weighting partially repairs the failure. We assign loss weights $\alpha = \{1.0, 1.0, 0.5, 0.25\}$ to the expert, first DAGger pool, second DAGger pool, and third DAGger pool respectively, preserving inverse-frequency token weights. The result is a large recovery over the naive three-step model: overall pass improves from 92.50% / 0.46% PV to 93.06% / 0.28% PV, Adversarial recovers from 87.92% / 0.42% PV to 89.17% / 0.00% PV, and Hidden recovers from 90.83% / 1.67% PV to 90.00% / 0.00% PV. Yet this still does not exceed the two-step frontier of 94.54% / 0.19% PV.

These ablations rule out the simplest stories. The three-step failure is not caused by a gross takeover of rare recovery tokens, and it is not fixed by removing class weighting. The evidence instead points to

Table 7: Nested token-count ablation over Gauntlet, Adversarial, and Compositional suites. Each row uses seeds 1–80; PV uses the same failed-rollout $C > 0.90$ criterion as Table 2. The full 9-token set removes observed PVs across the three suites, while reduced sets leave at least one suite with nonzero PVs.

Token set	Gauntlet		Adversarial		Compositional	
	Pass	PV	Pass	PV	Pass	PV
4-token	89.8%	0.6%	90.6%	0.6%	94.2%	0.2%
5-token	94.6%	0.0%	87.5%	0.6%	92.2%	0.2%
7-token	94.0%	0.0%	88.1%	0.0%	91.8%	0.2%
9-token	94.2%	0.0%	91.9%	0.0%	92.2%	0.0%

Table 8: Interaction-aware trajectory-scorer ablation on the same held-out LHS sweep. Entries are pass/PV percentages. Scorers evaluate candidate futures directly, but still trail two-step DAgger on targeted stress suites.

Policy	Gauntlet	Adversarial	Hidden
Interaction Scorer (learner states)	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67
Interaction Scorer (merged states)	88.75 / 1.67	82.92 / 2.50	90.00 / 2.50
Token-DAgger-BC (2-step)	96.25 / 0.28	90.83 / 0.00	91.67 / 0.00
Spotlight Reflex	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

a *late-pool aggregation pressure*: once overlapping nominal and recovery pools are mixed uniformly, later recovery states flatten the classifier’s boundary. Down-weighting those late pools mitigates the effect, but only partially, consistent with a calibration problem under limited model capacity rather than with a pure data-distribution shift.

Token-count coverage bound. Removing a response class creates a maneuver gap: STOP handles hard blockage within braking distance; CRAWL/MAINTAIN handle low/high-speed forward progress; SLOW_YIELD handles yield-without-stopping; NUDGE_{L/R} handles corridor bias without full lane departure; EVASIVE_{L/R} handles head-on and cut-in scenarios requiring larger lateral displacement; LANE_RECOVER handles post-evasion re-centering. Table 7 reports a nested token-count ablation. It supports the claim that the full token set closes observed PV gaps in this simulator, but it does not prove global optimality of nine tokens.

5.3 Trajectory-Scorer Ablation

A natural alternative to hard-coded scoring is to learn a utility function over the same candidate trajectories. We tested trajectory-informed scorers that evaluate each token’s short-horizon ego rollout, augmented with kinematic invariants (curvature, heading deltas, path length, lateral-acceleration and jerk proxies) and explicit interaction features (minimum dynamic clearance, time to closest approach, signed longitudinal/lateral miss distances, relative speed, and crossing relation). These models are trained contrastively: the Spotlight-selected candidate is the positive and the other token candidates are negatives.

Table 8 rules out a simpler explanation: the remaining gap is not solved by replacing the flat token head with a candidate utility scorer over ego trajectories plus lightweight interaction features. The scorer sees the candidate futures, but it is still an offline selector. In closed loop it trails the two-step DAgger policy on Gauntlet and Adversarial and does not preserve the zero-PV hidden result. This points to closed-loop state coverage as the dominant ingredient in the current simulator, rather than a need for more expressive per-candidate scalar scoring.

5.4 Failure Mode Taxonomy

The 28 Gauntlet non-passes by Spotlight Reflex (Table 2; 480 scenarios, 94.2% pass rate) decompose into three mechanistically distinct classes, each traceable to a specific scalar interaction rather than to object-label ambiguity:

Over-stopping under cumulative pressure (46% of failures). Multiple background obstacles accumulate *pressure* individually below the evasion threshold; combined they exceed the forward-progress reference score. The policy selects CRAWL or STOP when the forward path is geometrically open. Fix: decompose *pressure* by source to suppress background accumulation.

Slow re-entry after evasion (30% of failures). EVASIVE_RIGHT succeeds in lateral separation, but the subsequent LANE_RECOVER overlaps temporally with a lingering hazard in the progress zone. The progress gate penalizes re-entry, causing a stall. Fix: widen the progress gate temporal window post-evasion.

Synchronized hazard overload (24% of failures). Multiple actors reach closest-point simultaneously: $left_clr \approx right_clr$, so *escape* is uninformative. Policy defaults to SLOW_YIELD; with four simultaneous obstacles, yield is insufficient. Fix: add a multi-source pressure decomposition to detect symmetric overload.

In all three cases the failure is traceable to a scalar computation, not to an object label—consistent with the geometry-invariance property. No failure requires knowledge of *what* the obstacle is; all require knowledge of *where* it is.

6 Limitations and Future Work

2D abstract simulation. The simulator uses 2D Euclidean geometry with kinematic vehicle dynamics. The AlphaSim smoke test (Sec. 4.3) checks adapter feasibility, not transfer validity. A full physics simulation (e.g., CARLA, nuPlan with dynamics) or real-vehicle evaluation is needed before making vehicle-dynamics claims.

Token-count optimality is not established. Table 7 is a nested removal study, not a search over all token libraries. The stronger claim that nine is near-minimal requires random token libraries, learned token sets, parameter perturbations of offsets and speed scales, and larger token counts (e.g., 12-token variants) to test whether extra classes induce token-flip instability.

Stress-test operating envelope (empirical). To quantify the *operating boundary* of geometric reflex vs. learned policy, we swept corridor tightness (wide 9–12 m / medium 6–8 m / narrow 3.6–5 m / extreme 2.2–3.6 m) $\times$ obstacle-actor pressure (1.0 $\times$ / 1.5 $\times$ / 2.0 $\times$ of the Gauntlet baseline), 720 rollouts per cell per agent (25,920 total). Table 9 summarises the PV rates.

Table 9: Phase diagram: worst-case Proximity Violation (PV) rate per corridor level (max over 3 pressure settings, 720 rollouts each). Spotlight Reflex $\leq 0.4\%$ PV across all 12 conditions; BC peaks at 6.5% in narrow/extreme corridors. Full 12-cell results in the artifact.

Corridor	ContBC	TokBC	Spotlight	Δ Tok-SR
Wide (9–12 m)	4.3%	3.9%	0.4%	+3.9 pp
Medium (6–8 m)	4.2%	3.6%	0.1%	+3.5 pp
Narrow (3.6–5 m)	5.6%	6.5%	0.3%	+6.4 pp
Extreme (2.2–3.6 m)	4.4%	6.4%	0.3%	+6.4 pp

Spotlight Reflex holds $\leq 0.4\%$ PV rate in all 12 cells. BC agents degrade as corridors narrow: the Token-BC delta grows from +2.2 pp (wide, 2.0 $\times$) to +6.4 pp (narrow/extreme, 1.0 $\times$ /1.5 $\times$). BC agents were trained on compositional and adversarial suites with default wide-lane topologies (6–9 m

half-widths); the narrow and extreme cells push *left_clearance* and *right_clearance* outside the training distribution, and the learned boundary degrades. Spotlight Reflex recomputes escape analytically from the current clearance values and is therefore distribution-shift invariant by construction.

This phase diagram should be read as an operating-envelope visualization, not as the primary environment proof. The stronger evidence is the held-out Latin-hypercube sweep in Sec. 4.2, which independently samples generator parameters and includes recurrent BC plus DAgger-style on-policy correction.

Notably, Token-BC *exceeds* Continuous-BC at extreme OOD (4.4% vs 6.1–6.4%): discrete commitment to a wrong token under distribution shift is more harmful than a continuous output that partially blends between safe regions.

Temporal scalar history and joint interaction. The strongest learned model in this paper is still a token classifier over 10 scalar features. Token-RNN-BC adds scalar history but remains weak, and interaction-aware trajectory scorers do not close the stress-suite gap. A richer joint future encoder over ego and actor rollouts may be needed to distinguish coordinated interaction from static clearance geometry.

DAgger aggregation is not solved. Two-step DAgger is the strongest baseline, but three-step DAgger regresses. This suggests that blindly adding recovery-state data can distort the training distribution. Future work should study aggregation schedules, loss weighting by rollout source, validation splits by closed-loop profile, and safety-constrained selection among DAgger checkpoints.

7 Conclusion

We have argued that discretizing the driving action space into margin-separated maneuver classes can be a structural robustness prior when safe trajectory sets are disconnected. The non-interpolation theorem formalizes the condition: token selection avoids unsafe intermediate trajectories that continuous interpolation must traverse. The empirical result is more nuanced than the initial fixed-suite comparison. Plain BC fails even with the same tokens and geometric features, and recurrent BC remains weak under held-out generator shift. But two-step DAgger nearly closes the frontier, matching Spotlight Reflex overall and reducing PV rate on the held-out Latin-hypercube sweep.

The broader implication is that action-space structure, data distribution, and selector training must be studied together. Tokens remove unsafe interpolation paths, geometric rules provide an interpretable safety prior and expert relabeler, and on-policy correction can learn most of that rule. The remaining open problem is not whether tokens can be learned at all; it is how to train token selectors that retain the two-step DAgger safety frontier without brittle aggregation or simulator-specific tuning. The present evidence is simulator-bound; proving the principle as a general property of safe navigation requires physics simulation, real-vehicle validation, and broader scenario generation.

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